

Mihir Panchal

Vadodara, Gujarat, India | mihirpanchal616@gmail.com | +91-6351510619

[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Kaggle](#)

Profile Summary

AI & Backend Engineer with experience building production-ready AI agents, Retrieval-Augmented Generation (RAG) systems, and scalable backend applications. Skilled in Python, FastAPI, LangChain, LangGraph, LLM integration, and API development. Experienced in designing enterprise AI solutions that analyse organizational data, generate intelligent reports, and automate business workflows. Passionate about building scalable, secure, and intelligent AI systems that solve real-world business problems.

Professional Experience

Freelance

- AI Engineer** May 2026 – Present
 - Designed, developed, and maintained production-ready AI applications using Python, FastAPI, LangChain, LangGraph, and LangSmith.
 - Built scalable backend services and REST APIs to support AI agent workflows and enterprise applications.
 - Developed Retrieval-Augmented Generation (RAG) pipelines to analyse organizational knowledge and deliver accurate, context-aware responses.
 - Integrated Large Language Models (LLMs) with vector databases, document processing pipelines, and external APIs to enhance AI capabilities.
 - Optimized prompt engineering, agent orchestration, memory management, and workflow execution for improved accuracy, scalability, and performance.
 - Collaborated with clients to understand business requirements and continuously deliver secure, scalable, and high-quality AI solutions.

NetWeb Software, Vadodara

- AI Engineer** Jun 2025 – Apr 2026
 - Designed and deployed enterprise-grade Generative AI and Agentic AI solutions to automate business workflows.
 - Built AI agents using LangChain, LangGraph, and LLMs, integrating internal knowledge bases through Retrieval-Augmented Generation (RAG).
 - Developed scalable backend APIs and AI workflows using Python and FastAPI for production deployments.
 - Optimized prompt engineering, agent orchestration, and response quality to improve reliability and user experience.
 - Collaborated with cross-functional teams to integrate AI capabilities into enterprise products and deliver production-ready solutions.
- Software Developer** Jul 2024 – May 2025
 - Built scalable C#, React, and Python**-based applications supporting enterprise clients.
 - Developed secure APIs, optimized performance, and automated workflows.
 - Enhanced data visualization by implementing **Power BI dashboards**.
- Software Developer Intern** Jan 2024 – Jun 2024
 - Customized and automated **OpenAir NetSuite** workflows for client operations.
 - Improved backend logic and reporting features, reducing manual errors by 20%.

Asence Pharma Pvt. Ltd.

- Full Stack Developer Intern** May 2023 – Jul 2023
 - Developed **HRMS Recruitment and Employee Master modules**, streamlining HR processes.
 - Designed optimized databases for large-scale employee data, ensuring faster query responses.

• **Student Placement Coordinator**

Jun 2023 – May 2024

- Coordinated student placements, strengthened industry partnerships, and supported career development initiatives.
- Facilitated workshops and technical training sessions for students.

Key Projects

Nexus AI – Multi-Domain Recommendation Engine

An intelligent, AI-powered recommendation system delivering hyper-personalized suggestions across Movies, Books, Video Games, Tech, and Travel. Uses advanced reasoning to match user vibes, preferences, and deal-breakers rather than basic genre-based filtering.

Generative AI · Machine Learning · React · TypeScript · Google Gemini · Prompt Engineering

DocuChat AI – Intelligent Document Assistant

A high-performance AI document workspace that enables real-time interaction with PDFs and text files. Supports contextual question answering with citations, smart summarization, and a streaming chat UI for accurate, human-like document analysis.

Generative AI · RAG · Prompt Engineering · React 19 · TypeScript · Tailwind CSS · Google Gemini

YOLO Vision Pro – Real-Time Object Detection

A browser-based computer vision application performing real-time object detection entirely on-device. Built with TensorFlow.js and COCO-SSD for low-latency inference, ensuring offline capability and complete user privacy.

Computer Vision · TensorFlow.js · YOLO · React · TypeScript · HTML5

Skills

- **AI/ML & Data Science:** Generative AI, Agentic AI, NLP, Machine Learning, RAG, LangChain, LangGraph, LangSmith, Prompt Engineering, Vector Databases
- **Programming & Backend:** Python, FastAPI, C#, React, JavaScript, TypeScript, SQL, Flask
- **Cloud & Tools:** AWS (EC2, S3, RDS, SageMaker), Power BI, Git, Streamlit, OpenAir Application

Certifications

- **Oracle AI Foundations Associate (2025):** Validated expertise in AI concepts and Oracle Cloud Infrastructure, focusing on enterprise solutions.
- **AWS Machine Learning & NLP:** Gained practical knowledge of ML algorithms, supervised/unsupervised learning, and NLP model deployment using AWS SageMaker.
- **AWS Cloud Foundations:** Acquired strong understanding of cloud computing fundamentals, AWS services, and best practices for cloud-native development.
- **Google – Generative AI, LLMs, Responsible AI:** Learned advanced prompt engineering, LLM fundamentals, and ethical AI deployment strategies.
- **HackerRank – Python, Pandas, SQL (Gold Badge):** Recognized for problem-solving skills, coding challenges, and advanced data manipulation techniques.

Education

Navrachana University, Vadodara – B.Tech, Computer Science & Engineering (2020 – 2024)

Sarwa Mangal School, Vadodara – Higher Secondary Education – Science (PCM) (2018 – 2020)

Achievements & Interests

- Completed multiple AI & ML certifications, including Responsible AI & Generative AI.
- Active contributor on Kaggle & GitHub with AI-focused projects.
- **Interests:** AI ethics, geospatial analytics, cloud solutions, and problem-solving.